

GUTHRIE, OK, USA

WMO: 723537

Lat: 35.852N

Lon: 97.414W

Elev: 326

StdP: 97.48

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53913

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.2	-7.5	-16.9	0.9	-7.9	-14.0	1.2	-5.2	11.4	13.5	10.6	12.7	2.6	0	0.516

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	38.9	23.2	37.4	23.7	35.8	23.8	25.7	33.6	25.2	33.1	24.7	32.3	4.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.7	19.2	29.6	22.9	18.3	28.8	22.5	17.9	28.3	81.3	33.6	78.8	33.2	76.4	32.4	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.8	9.3	8.4	-15.0	41.0	2.9	2.1	-17.1	42.5	-18.8	43.7	-20.5	44.9	-22.6	46.5
			-15.6	27.2	2.8	1.2	-17.6	28.1	-19.2	28.8	-20.8	29.4	-22.8	30.3
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.5	4.0	5.7	11.5	16.3	21.0	26.0	28.7	28.3	23.8	17.1	10.7	4.8
	DBStd	10.07	5.97	6.41	6.11	5.16	4.44	3.16	3.16	3.33	4.07	5.25	5.58	5.67
	HDD10.0	655	202	148	54	9	1	0	0	0	0	8	57	176
	HDD18.3	1891	446	354	223	97	25	1	0	0	6	87	233	419
	CDD10.0	3044	15	27	101	199	342	478	579	566	415	227	79	16
	CDD18.3	1240	0	1	12	37	109	229	321	307	171	48	6	1
	CDH23.3	14079	2	10	92	334	1028	2511	4096	3786	1749	427	43	1
	CDH26.7	6982	0	2	17	89	367	1180	2302	2107	785	127	5	0
Wind	WSAvg	3.9	3.7	3.9	4.4	4.8	4.3	4.3	3.7	3.5	3.4	3.7	3.9	3.7
Precipitation	PrecAvg	940	34	43	74	77	136	121	71	74	111	82	65	46
	PrecMax	1337	101	173	317	188	303	309	241	234	329	271	236	169
	PrecMin	578	0	0	2	4	16	18	0	0	5	0	0	0
	PrecStd	199	29	35	59	46	72	68	55	55	78	62	53	38
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.9	24.7	28.8	32.2	34.2	38.4	41.5	41.9	38.5	32.3	27.3	22.0
		MCWB	12.0	14.8	16.9	18.9	21.8	23.3	22.8	23.5	22.2	20.5	17.3	14.2
	2%	DB	19.1	21.9	26.0	29.0	32.4	35.9	39.0	38.8	34.8	29.9	24.1	18.8
		MCWB	10.9	13.3	15.9	18.0	21.9	23.7	23.2	23.4	22.8	20.1	15.7	13.2
	5%	DB	16.8	18.8	23.4	27.2	30.8	33.9	37.4	37.4	32.9	27.7	22.2	16.3
		MCWB	9.9	12.1	15.2	17.5	21.7	23.4	23.9	23.8	22.5	18.6	14.9	10.8
	10%	DB	13.6	16.3	21.3	25.0	28.5	32.5	35.8	35.6	31.2	25.8	19.6	13.9
		MCWB	7.6	10.2	14.3	16.9	20.8	23.4	24.0	23.9	21.7	18.3	13.7	8.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.6	16.9	19.3	21.6	24.7	26.3	26.7	26.5	25.0	23.2	19.6	17.3
		MCDB	18.3	21.1	25.9	27.8	31.1	33.4	34.0	35.0	32.5	29.0	23.7	20.7
	2%	WB	12.5	15.2	18.0	20.3	23.6	25.4	25.8	25.6	24.1	22.0	17.5	14.3
		MCDB	17.4	19.4	23.2	26.1	29.7	32.6	33.5	34.0	30.9	26.6	21.7	16.9
	5%	WB	10.1	13.2	16.9	19.5	22.8	24.7	25.3	25.1	23.5	20.9	16.3	11.5
		MCDB	15.7	18.0	21.8	24.8	28.5	31.6	33.0	33.4	30.1	25.3	20.3	15.2
	10%	WB	8.0	10.4	15.3	18.4	21.8	24.0	24.7	24.4	22.8	19.5	14.9	9.0
		MCDB	12.8	16.0	20.0	23.4	27.1	30.5	32.4	32.4	29.2	24.1	19.0	13.2
Mean Daily Temperature Range	5% DB	MDBR	11.9	11.8	12.4	12.2	11.2	10.8	11.6	11.9	12.0	12.2	12.0	11.1
		MCDBR	16.1	15.4	14.4	14.2	12.7	12.2	13.5	14.0	13.0	13.9	14.1	14.3
		MCWBR	9.6	9.2	7.8	7.0	5.6	4.4	3.9	4.2	4.6	6.6	7.7	9.2
	5% WB	MCDBR	14.4	13.4	11.7	11.8	11.1	10.4	11.5	11.7	11.0	10.8	11.6	12.4
		MCWBR	9.7	9.4	7.2	7.2	5.9	4.7	4.2	4.4	4.4	6.2	8.0	9.9
Clear-Sky Solar Irradiance	taub	0.296	0.300	0.340	0.380	0.418	0.426	0.429	0.437	0.390	0.348	0.312	0.290	
	taud	2.496	2.501	2.410	2.314	2.281	2.300	2.323	2.247	2.359	2.465	2.502	2.521	
	Ebn at Noon	909	947	929	903	869	859	852	839	865	878	881	895	
	Edh at Noon	82	91	110	127	134	131	127	135	114	94	81	75	
All-Sky Solar Radiation	RadAvg	2.73	3.35	4.38	5.44	5.93	6.77	6.80	6.17	5.11	3.86	2.89	2.30	
	RadStd	0.27	0.50	0.28	0.39	0.62	0.61	0.37	0.37	0.43	0.48	0.37	0.25	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (37 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page